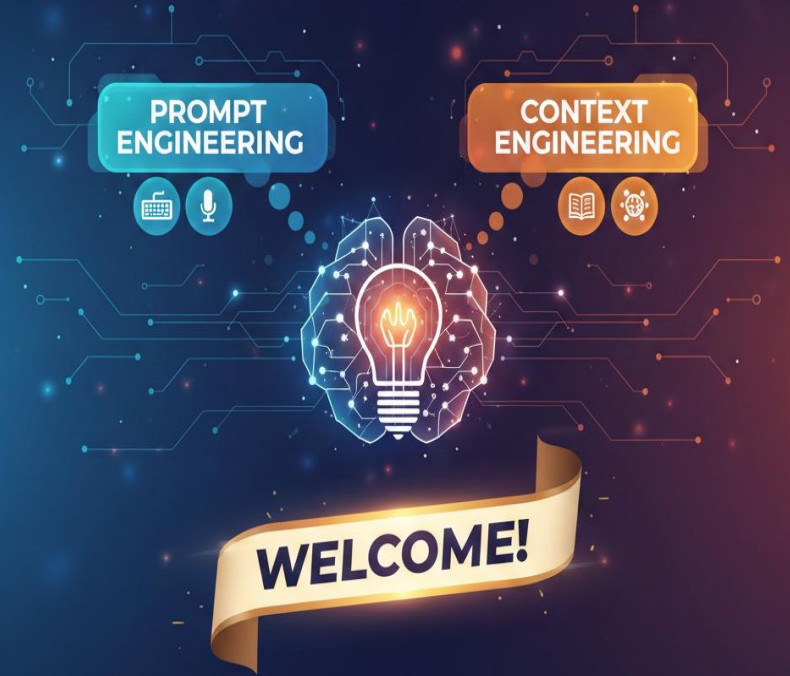


Prompt Engineering and Context Engineering



What is Prompt Engineering?

- Prompt engineering is the art and science of crafting instructions that guide AI language models to produce desired outputs.
- Think of it as learning to communicate effectively with AI systems to achieve specific goals.
- **Prompt:** A prompt is the text or instruction you give to an AI model so it knows what you want.
- Example: “Write a poem about rain in Urdu.”



Components of a Good Prompt

Persona

- Who the AI should act as.

Task

- What you want the AI to do.

Context

- Background information and relevant details.

Format

- How you want the response delivered.

Components of a Good Prompt

➤ Persona:

- Who the AI should act as.
- Example-1: "You are a business man"
- Example-2: "You are a student of CS"
- Example-3: "You are an Mechanical Engineer"



➤ Task:

- What you want the AI to do.
- Always includes a verb or command,
- Example-1: "Draft a business proposal.."
- Example-2: "Write an application.."
- Example-3: "Update this Project requirement.."



Components of a Good Prompt?

➤ Context:

- Background information and relevant details.
- Example-1: "Based on details about relevant Documents..."
- Example-2: "Based on Python Programming ..."
- Example-3: Based on thermodynamics...

➤ Format:

- How you want the response delivered.
- Example-1: "Limit to bullet points."
- Example-2: "In the form of Table."
- Example-3: "In pdf document"



Good Prompt Examples

➤ **Example-1:**

You are a Senior Software Architect. Draft a technical project assessment for a Computer Science student preparing a capstone proposal. The project involves building a real-time web chat application using Python, Web Sockets, and Redis. The student has implemented the backend logic, the core unit tests have passed, performance benchmarking is still pending, the project advisor has approved the initial architecture, and the project deadline is in 14 days. Present the assessment in bullet points under the headings Architecture Overview, System Strengths, Identified Bottlenecks, Technical Questions for Review, Recommended Frameworks, and Next Action Items.

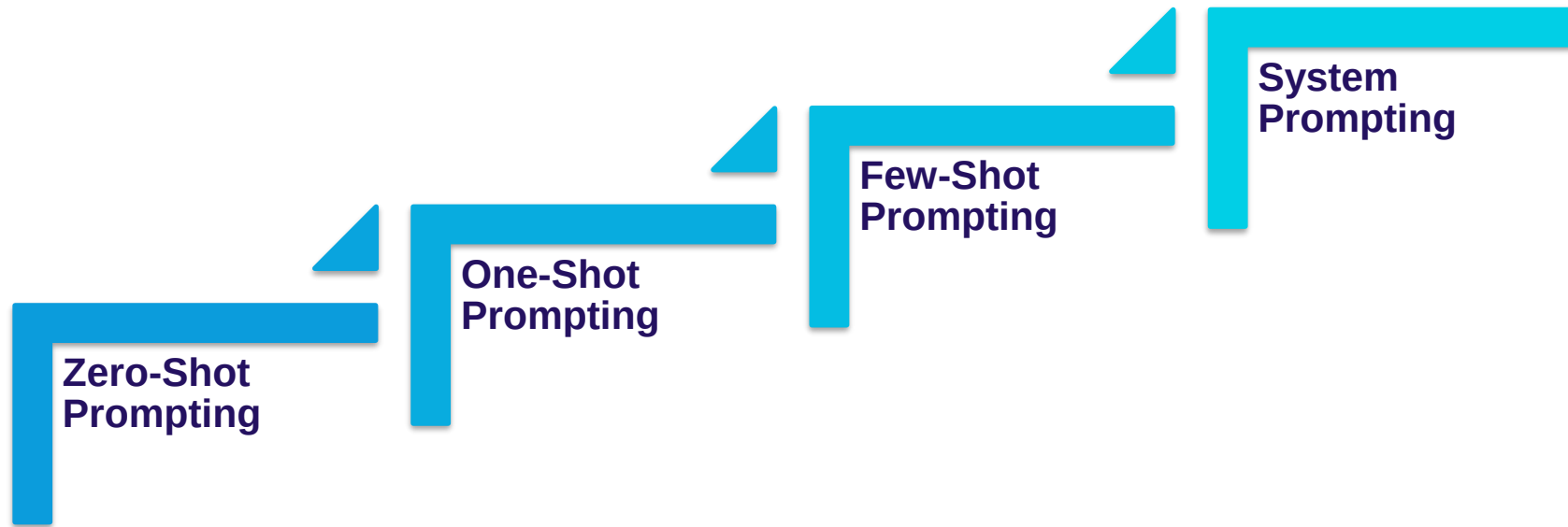
Good Prompt Examples

➤ Example-1:

You are a Senior Software Architect. Draft a technical project assessment for a Computer Science student preparing a capstone proposal. The project involves building a real-time web chat application using Python, Web Sockets, and Redis. The student has implemented the backend logic, the core unit tests have passed, performance benchmarking is still pending, the project advisor has approved the initial architecture, and the project deadline is in 14 days. Present the assessment in bullet points under the headings Architecture Overview, System Strengths, Identified Bottlenecks, Technical Questions for Review, Recommended Frameworks, and Next Action Items.



Fundamental Prompting Techniques



Zero-Shot Prompting

- You just tell it what to do.
 - **Prompt:** Draft a Python function to reverse a singly linked list. Use clean, PEP 8 compliant code.
 - **Prompt:** Draft a brief executive summary proposing a strategic partnership with a local distributor. Use a professional corporate tone.
 - **Prompt:** Generate inspection checklist questions for a plant supervisor conducting a preventative maintenance check on a centrifugal pump.

One-Shot Prompting

- You give one sample to show the model what style you want.
 - **Prompt:** Using the same tone, structure, and level of detail, draft a professional response to a customer refund request for Order No. 8842.
 - **Example-1:** "Regarding Order No. 5120, the client reports that the item arrived with minor surface damage due to transit. Courier logs confirm a delayed delivery, and the standard 14-day return window remains valid. As the product is eligible for partial reimbursement and photo proof has been submitted, the client is approved for a 20% partial refund."

Few-Shot Prompting

- Instead of explaining, show the template how the task should look.
 - **Prompt:** Using the following style and tone, draft a bug triage summary for Issue No. 405.
 - **Example-1:** Issue No. 102, the user reports a runtime crash during file upload. The lead developer and QA tester are investigating, and a fix targeting the buffer size has been merged to dev.
 - **Example-2:** In Issue No. 215, the student group reports an unhandled API exception in the login module. The missing parameter validation has been patched, and integration tests are currently running.
 - **Example-3:** In Issue No. 308, the peer reviewer noted memory leaks in the sorting function. The profiler logs confirm high consumption, and the team has already scheduled a code refactor.

System Prompting

- Set overall context and behavior guidelines.
 - **Prompt:** You are assisting a Chief Executive Officer. All outputs must follow formal executive summary formatting, include key performance indicators (KPIs) where appropriate, and avoid speculative financial forecasts. Do not fabricate market figures, revenue data, or corporate policies.
 - **Prompt :** You are assisting a senior software architecture instructor. All outputs must follow formal technical writing standards, include complexity notations (Big-O) where appropriate, and avoid vague performance claims. Do not fabricate libraries, syntax, or API references.
 - **Prompt:** You are an AI corporate strategy advisor for executive management. Maintain a professional, concise, and business-focused tone. Base your analysis strictly on financial metrics, operational risks, and market data. Always follow standard corporate reporting and strategic planning standards.

Chain-of-Thought Prompting

- Tell the AI to think step-by-step, just like an executive decision-making process.
 - **Prompt:** Analyze whether acquiring Company X is financially justified. Do not jump to the conclusion. First evaluate:
 1. Operational synergies and cost savings
 2. Valuation and debt-to-equity ratio
 3. Market penetration risks

Give reasoning first, then give the final view..

Role Prompting

- Tell the model who it must act as.
 - **Prompt:** Act are a Sessions Judge, prepare a written judgment in a narcotics case. Maintain judicial neutrality and adhere strictly to statutory language..
 - Why it works: Judges get a response written in their professional voice, not some random chatbot poetry.





CONTEXT ENGINEERING IN AI



What is Context Engineering?

- Context Engineering is the practice of designing and building dynamic systems that provide a Large Language Model (LLM) with the right information, in the right format, at the right time to accomplish a specific task.
- In simpler terms, you're strategically "packing" the context window (the input area of an LLM) to maximize effectiveness. Think of it as creating the perfect instruction manual for your AI system.
- *Think of the AI model as a judge's mind, and the context window as the case file placed before the judge.*

Prompt Vs. Context Engineering

Perspectives	Prompt Engineering	Context Engineering
Goal	Tell the model how to behave and what to produce	Give the model the facts/examples it should rely on
Levers	Wording, structure, roles, constraints, output schema, few-shot examples	Retrieval (RAG), documents, knowledge bases, tools/APIs, memory, state across turns
Typical changes	“Be concise. Return valid JSON with fields X/Y/Z.”	“Attach the company glossary, latest policy PDF, and retrieved passages for this query.”
Failure mode	Vague instructions → messy/incorrect format	Missing/irrelevant info → hallucinations/outdated answers
Ownership	UX/prompt designers, app devs	Data/ML/platform teams (pipelines, indexing, chunking, evals)